

Review from previously:

1. **Definition:** A quadratic form on $\mathbb{R}^n$ is a function Q whose value at x can be computed using the rule $Q(x) = x^T A x = x \cdot A x$ where A is a symmetric matrix.
2. Minkowski 'inner product'. Write $\mathbb{R}^{n+1}$ as (t, y) with $y \in \mathbb{R}^n$.
 - (a) $Q((t, y)) = \|y\|^2 - t^2$.
 - (b) The theory of special relativity is all about this Minkowski inner product.
3. Change of variable
 - (a) If $x = Py$ where P is invertible, then $x^T A x = y^T P^T A P y$.
 - (b) Thus, in the new coordinates, Q is represented not by A , but by $P^T A P$.
4. Principle axis theorem: There is a change of variable matrix P that is orthonormal such that $Q(y) = y^T D y$ with D being diagonal (the eigenvalues of A are its diagonal entries).
5. Constrained optimization: Look at $Q(x)$ subject to the constraint that $\|x\| = 1$.
 - (a) If A is the symmetric matrix such that $Q(x) = x^T A x$, then the maximum of Q on the constraint surface is the largest eigenvalue of A ; and likewise the minimum is the smallest eigenvalue of A .
 - (b) Minimize Q on the $\|x\| = 1$ sphere subject to the constraint that x be orthogonal to all eigenvectors previously found.

1. We looked at symmetric matrices: Let A be a symmetric $n \times n$ matrix.
 - (a) The matrix A is orthogonally diagonalizable: $A = PDP^{-1}$ with $P^{-1} = P^T$.
 - (b) The columns of P are an orthonormal basis $\{v_1, \dots, v_n\}$ of eigenvectors for A .
 - (c) The eigenvalues of A are real and are the diagonal entries of D .
2. What can be said about an $m \times n$ matrix A when $m \neq n$?
 - (a) $A^T A$ is a symmetric, $n \times n$ matrix. (We used this matrix $A^T A$ when doing least squares and orthogonal projection to a subspace in $\mathbb{R}^n$.)
 - (b) Thus, it is orthogonally diagonalizable: $A^T A = PDP^{-1}$ with $P^{-1} = P^T$.
 - (c) The eigenvalues are non-negative because $v \cdot A^T A v = (Av) \cdot (Av) = \|Av\|^2$.
 - (d) The square roots of the eigenvalues $\sigma_1 = \sqrt{\lambda_1}, \dots, \sigma_n = \sqrt{\lambda_n}$ are called the singular values.
 - (e) $\text{Null}(A)$ has dimension at least $n - m$, so at most m of the λ_k 's are not zero.

3. We can likewise look at AA^T which is an $m \times m$ symmetric matrix.
 - (a) It can be written as QBQ^{-1} where Q is an $m \times m$ orthogonal matrix and B is diagonal with entries being the eigenvalues of AA^T .

4. **Theorem:** The non-zero eigenvalues of AA^T are the same as those of $A^T A$.
 - (a) Indeed, if $A^T Av = \lambda v$, then $A(A^T Av) = \lambda Av$ and $A(A^T Av)$ is $(AA^T)Av$.

 - (b) Also, if $AA^T u = \lambda' u$, then $A^T(AA^T u) = \lambda' A^T u$.

 - (c) So if v_k is an eigenvector of $A^T A$ with non-zero eigenvalue λ_k and length 1, then $u_k = \lambda_k^{-1/2} Av_k$ is an eigenvector of AA^T with eigenvalue λ_k and length 1.

 - (d) Moreover $u_k \cdot u_j = \lambda_k^{-1/2} \lambda_j^{-1/2} (Av_k) \cdot (Av_j) = \lambda_k^{-1/2} \lambda_j^{1/2} v_k \cdot A^T Av_j = \lambda_k^{-1/2} \lambda_j^{1/2} v_k \cdot v_j$.

 - (e) Thus, the $\{u_k = \lambda_k^{-1/2} Av_k : \lambda_k \neq 0\}$ are an orthonormal basis for the eigenspaces of AA^T with non-zero eigenvalue.

5. Thus, AA^T and $A^T A$ have the same set of non-zero eigenvalues (which are non-negative). List these as $\lambda_1, \dots, \lambda_r$ for $r \leq \min(m, n)$.

6. We can then write $A = \sigma_1 u_1 v_1^T + \dots + \sigma_r u_r v_r^T$ where $\{v_i\}_{i=1, \dots, r}$ are an orthonormal basis for the eigenspaces of A with non-zero eigenvalue, and $u_k = \sigma_k^{-1} A v_k$.

(a) Indeed, if x is any vector in $\mathbb{R}^n$, I can use the basis $\{v_1, \dots, v_r, v_{r+1}, \dots, v_n\}$ to write x as $x = c_1 v_1 + \dots + c_n v_n$.

(b) Then $Ax = \sigma_1 c_1 u_1 + \dots + \sigma_r c_r u_r$ which gives the expansion of Ax in terms of the u -basis. (Note that if $AA^T u = 0$, then $A^T u = 0$ so Ax has no u_k 's in its expansion from the 0 eigenvalues of AA^T .) Likewise, there are no c_k for those v_k with Av_k being 0.

(c) In this regard, $u_k \cdot Ax = (A^T u_k) \cdot x = \sigma_k v_k \cdot x = \sigma_k c_k$ as it should be.

7. The depiction of A as $\sigma_1 u_1 v_1^T + \dots + \sigma_r u_r v_r^T$ is called the singular value decomposition of A .

1. We started with linear equations, wrote them in terms of matrices ($Ax = b$) and then introduced the row-reduction algorithm to solve them or find them inconsistent by obtaining $\text{rref}(A)$.
2. We then started playing with the notion of a matrix:
 - (a) Multiplying them against vectors, multiplying them against other matrices.
 - (b) The notion of an invertible matrix and its inverse.
 - (c) The set of invertible matrices form a group denoted by $GL(n; \mathbb{R})$.
 - (d) Introduced elementary matrices to formalize the row reduction operation.

Then, we discovered the determinant of a matrix which gives us a way to determine when a matrix is invertible.

3.
 - (a) Lots of neat properties of determinants ($\det(AB) = \det(A)\det(B)$).
 - (b) An inductive formula for the determinant using expansion across any row or column.
4. Introduced the notion of a vector space:
 - (a) Subspaces in $\mathbb{R}^n$.
 - (b) Null space and column space of a matrix.
 - (c) The notion of a basis for a vector space giving coordinates for the vector space, which allows us to write linear transformations as matrix multiplication.
 - (d) Change of basis formulas.
 - (e) Notion of dimension of a subspace and rank of a matrix (dimension of its column space).
5. The notion of an eigenvector and eigenvalue for a square matrix:
 - (a) Characteristic equation for potential eigenvalues.
 - (b) Diagonalizable matrices (not all are diagonalizable over $\mathbb{R}$).
 - (c) Complex matrices and complex eigenvalues (most matrices are diagonalizable if complex eigenvalues are allowed).
 - (d) Using diagonalization to solve discrete dynamical systems and linear differential equations.
6. Introduced dot products (inner products) and studied the many things it brings with it:
 - (a) Notion of length, distance, orthogonality.
 - (b) Orthogonal basis, orthonormal basis, orthogonal projections.
 - (c) Gram-Schmidt algorithm.
 - (d) Least squares and finding the closest vector in a subspace.
7. Continuing with dot products, we studied symmetric matrices:
 - (a) These are orthogonally diagonalizable (and vice versa).

- (b) There is an orthonormal basis consisting of eigenvectors. Eigenvalues are real.
- (c) Spectral theorem.
- (d) Singular value decomposition for $m \times n$ matrices even when $m \neq n$.

And we learned how to write proofs and interpret rigorous mathematics.

And we learned something about science and data.